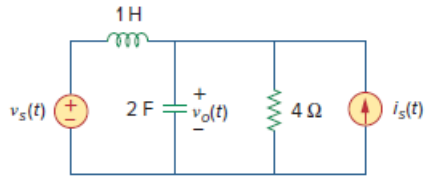


Problem

Develop the state equations for the circuit shown in Fig. 16.103.

Figure 16.103:



Step-by-step solution

Step 1 of 6

Refer to Figure 16.103 in textbook.

Draw the circuit as shown in Figure 1.

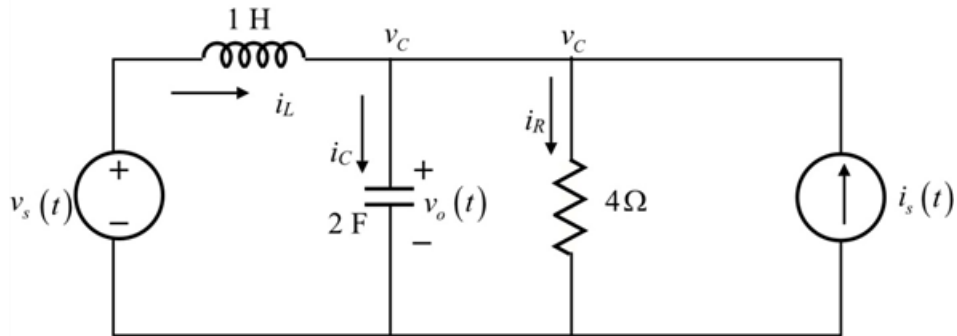


Figure 1

Step 2 of 6

Apply Nodal analysis at node v_C in Figure 1.

$$i_L = i_C + i_R$$

$$i_L = C \frac{dv_C}{dt} + \left(\frac{v_C}{R} + i_s \right)$$

$$C \frac{dv_C}{dt} = i_L - \left(\frac{v_C}{R} + i_s \right)$$

$$\frac{dv_C}{dt} = \frac{i_L}{C} - \frac{\left(\frac{v_C}{R} + i_s \right)}{C} \dots\dots (1)$$

Step 3 of 6

Consider,

$$\frac{dv_C}{dt} = v_C'$$

Replace 2 F for C , $4\ \Omega$ for R in equation 1.

$$v_C' = \frac{i_L}{2} - \frac{\left(\frac{v_C}{4} - i_s\right)}{2}$$

$$v_C' = \frac{i_L}{2} - \frac{v_C}{4(2)} + \frac{i_s}{2}$$

$$v_C' = 0.5i_L - 0.125v_C + 0.5i_s \dots\dots (2)$$

Step 4 of 6

Apply Kirchhoff's voltage law in loop1,

$$v_s = v_L + v_C$$

$$v_s = L \frac{di_L}{dt} + v_C$$

$$\frac{di_L}{dt} = \frac{v_s}{L} - \frac{v_C}{L}$$

$$i_L' = \frac{v_s}{L} - \frac{v_C}{L}$$

Replace 1 H for L in equation.

$$i_L' = v_s - v_C \dots\dots (3)$$

Step 5 of 6

Arrange equation 2 and equation 3 in state equation model.

$$\begin{bmatrix} v_C' \\ i_L' \end{bmatrix} = \begin{bmatrix} -0.125 & 0.5 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Therefore, the equation for the circuit at input is

$$\begin{bmatrix} v_C' \\ i_L' \end{bmatrix} = \begin{bmatrix} -0.125 & 0.5 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_s \\ i_s \end{bmatrix}.$$

Step 6 of 6

From Figure 1, $v_o(t)$ is the voltage across capacitor is same as the voltage across the at node

$$v_C.$$

$$v_o(t) = v_C \dots\dots (4)$$

Write the state equation model for equation 4.

$$v_o(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Therefore, the state equation is

$$v_o(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_s \\ i_s \end{bmatrix}.$$